UBICOMP ISWC2025

14-16 Oct. 2025
Helsinki, Finland

Visual Localization: GeoPoseProtocol and OpenVPS

Dr. Gábor Sörös
Nokia Bell Labs & Open AR Cloud

OPEN AR CLOUD

Vision: Ubiquitous, geographically anchored digital information

Location-based AR becomes mainstream



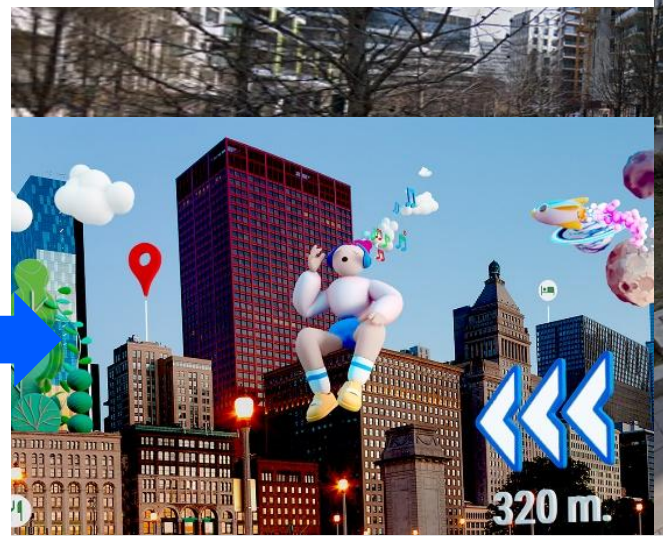
Niantic Pokémon GO, 2016



Spatial Internet, 2018
(concept by Judith Hecktor)

NIANTIC SPATIAL

Niantic Spatial, 2025



Google Geospatial AR, 2024



Google Street View at Corvin Budapest, 2022

Such physical hyperlinks make our world clickable!

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Spatial anchors and shared maps of the world

Required for re-identification of physical locations

- Anchoring 3D virtual assets to the real world requires precise and robust localization (<5cm, <1 deg)
- Visual localization provides this accuracy and enables recognizing the same location over time and across users
- Spatial anchor = localization + discovery + persistence
- Shared world map and shared contents stored in the “AR Cloud”
- Key players: Google, Apple, Meta, Niantic, Microsoft

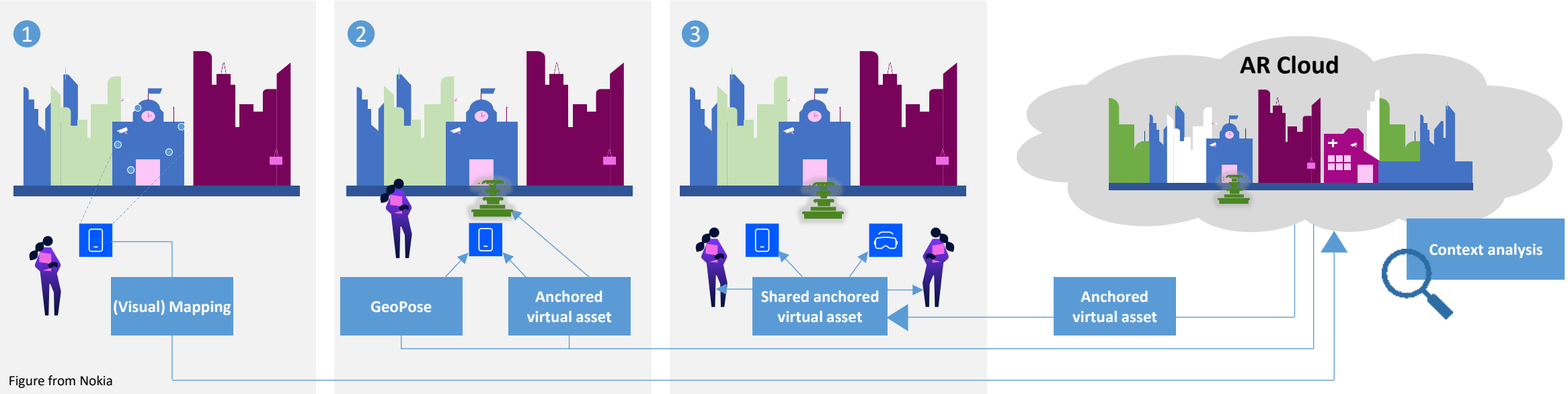
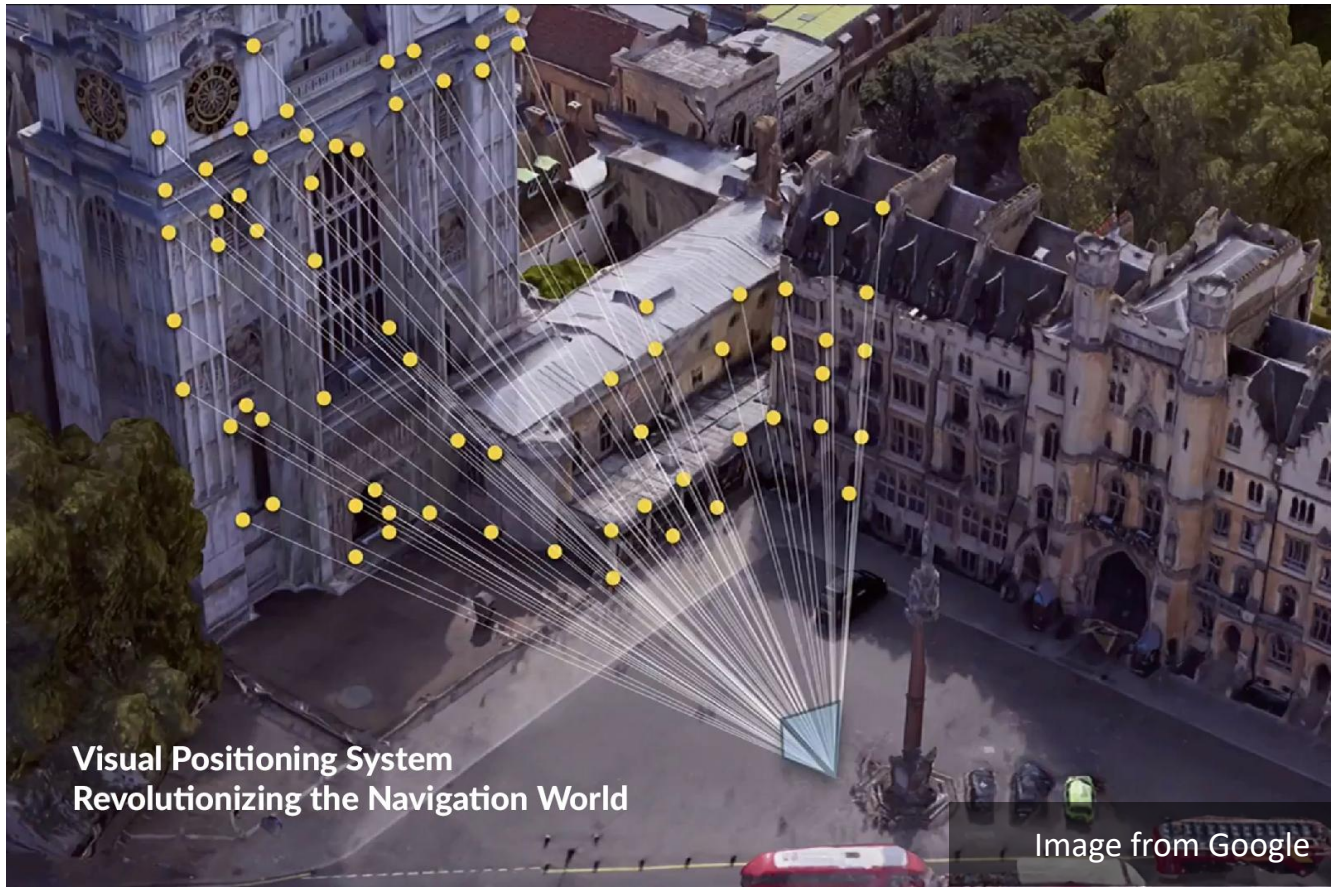


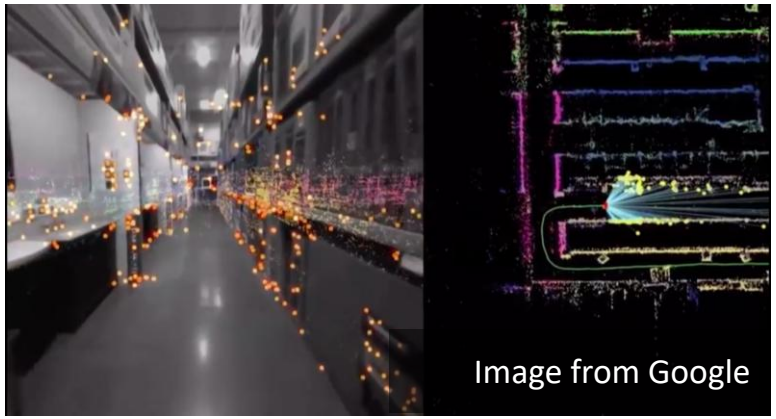
Figure from Nokia

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Visual positioning systems (VPS)

Camera-based precise 6DoF localization





Key players:

Google Visual Positioning
Apple GeoAnchors
Facebook (Scape, Mapillary, etc.)
Niantic (6D.ai)
(Microsoft Azure Spatial Anchors)
Augmented City
Immersal

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Scalable localization on mobile devices

Local visual-inertial odometry (VIO) + global visual positioning service (VPS)

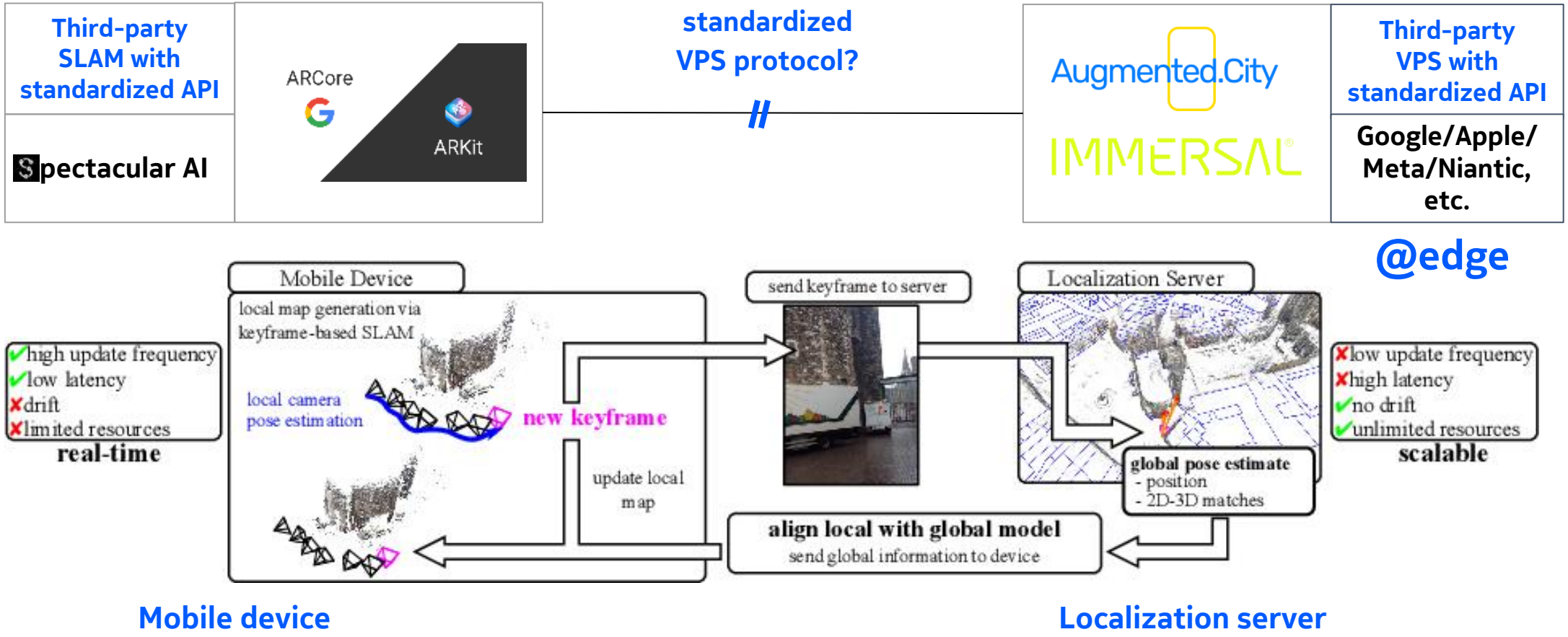


Figure adapted from Middelberg, Sattler, Untzelmann, Kobbelt - Scalable 6-DoF Localization on Mobile Devices, ECCV 2014

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GeoPoseProtocol (GPP)

https://github.com/OpenArCloud/oscp-geopose-protocol

A protocol proposal for visual positioning:

- send an image (and optionally other sensor data)
- receive the camera's 6DoF GeoPose

GeoPoseResponse (v2)

```
"id": str(uuid),
"timestamp": number,
"accuracy": GeoPoseAccuracy,
"type": "geopose",
"geopose": {
  "position": {
    "h": 0.034691,
    "lat": 30.272449,
    "lon": 59.934122
  },
  "quaternion": {
    "x":  0.0160586,
    "y": -0.2691833,
    "z": -0.0054743,
    "w": -0.9629394
  }
}
```

global location

ENU orientation

GeoPoseRequest (v2)

```
"id": str(uuid),
"timestamp": number,
"type": "geopose",
"sensors": [
  {"id": "0","type": "camera"},
  {"id": "1", "type": "geolocation"}
],
"priorPoses": GeoPoseResp[]?;
"sensorReadings": {
  "cameraReadings": [{
            "timestamp": number,
            "sensorId": "0",
            "reading": {
                "sequenceNumber": 0,
                "imageFormat": "JPG",
                "imageOrientation": {"mirrored": False,
                                     "rotation": 0},
                "imageBytes": img_file_base64_string
  }],
  "geolocationReadings":[{
            "timestamp": number,
            "sensorId": "1",
            "reading": {
                "latitude": float(lat),
                "longitude": float(lon),
                "altitude": 0
            }
  }],
  "wiFiReadings": WiFiReading[]?,
  "bluetoothReadings": BluetoothReading[]?,
  "accelerometerReadings": AccelerometerReading[]?,
  "gyroscopeReadings": GyroscopeReading[]?,
  "magnetometerReadings": MagnetometerReading[]?
}
```

image

GPS

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GeoPoseProtocol implementations

Protocol definition by Open AR Cloud:

https://github.com/OpenArCloud/oscp-geopose-protocol

JavaScript client implementation (v1) by Michael Vogt:

https://github.com/OpenArCloud/gpp-access

C++ client and server API (v2) by Gábor Sörös:

https://github.com/OpenArCloud/oscp-geopose-protocol/tree/master/cpp

Python client and server API (v2) by Gábor Sörös:

https://github.com/OpenArCloud/oscp-geopose-protocol/tree/master/python

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Minimal client and server

python/

- demo_client.py
- demo_server.py

cpp/oscp-gpp-demo/

- main_client.cpp
- main_server.cpp

data/

- seattle.jpg



- seattle_camera_params.json

request

```
{
    "camera_model": "OPENCV",
    "camera_params": [1499.027, 1499.027, 954.783, 511.957, 0.0, 0.0, 0.0, 0.0],
    "camera_id": "camera_01"
}
```

- seattle_geolocation_params.json

```
{
    "lat": 47.61155,
    "lon": -122.337056,
    "h": 42.0
}
```

response

- seattle_vps.json

```
{
    "geopose": {
        "position": { … },
        "quaternion": { … }
    }
}
```


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Nokia OpenVPS

Team



Asmaa Thabet
(engineering intern)



Andor Kovács
(engineering intern)



Dr. Dénes Bisztray
(sr. research engineer)



Balázs Ladjánszki
(freelancer)



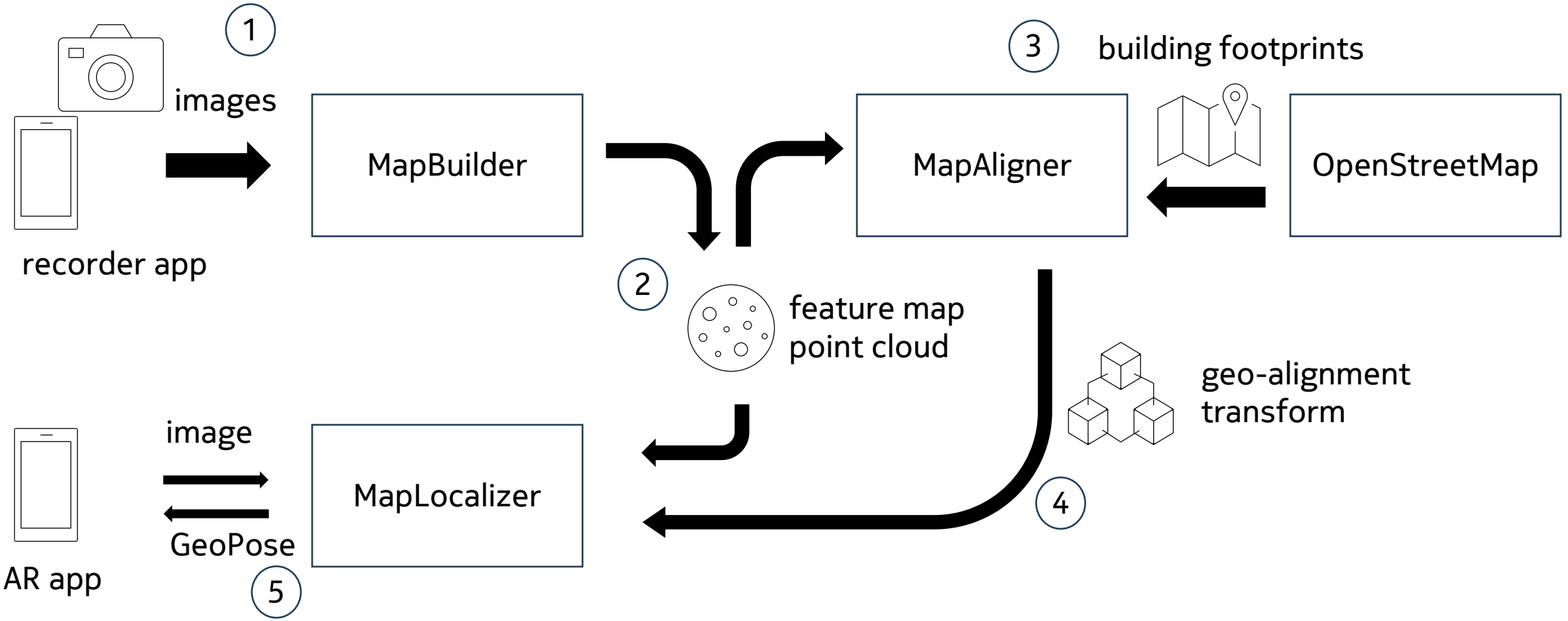
Dr. Gábor Sörös
(sr. research scientist)

Funded by:

OSCAR4US: Open-source, Scalable and Continuous Augmented Reality Services for Urban Spaces, 2025

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An Open Visual Positioning Service: Overview



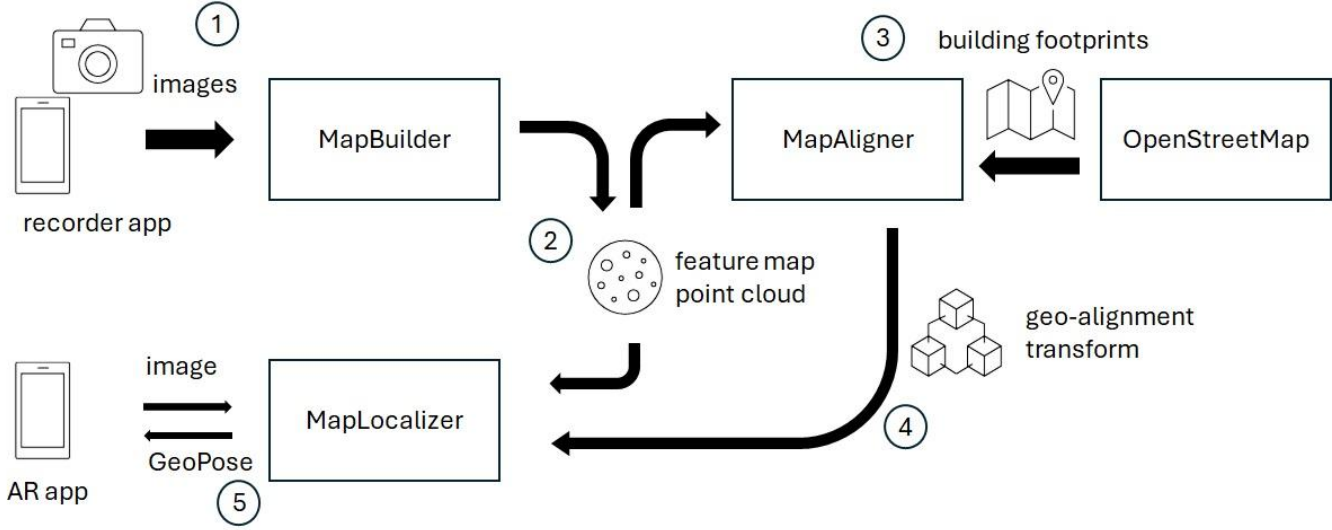
https://github.com/nokia/openarcloud-openvps

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Nokia OpenVPS

Visual positioning service for self-hosting

`https://github.com/nokia/openarcloud-openvps`









① Input photos

② OpenVPS point cloud map

③ Geo-aligned map in MapAligner

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Photo capture app

- Instead of implementing a new app, we found the open-source **StrayScanner** to be fit for our purposes

  https://github.com/strayrobots/scanner

- Collects RGB + depth + ARKit poses

- The user can zip the recording on the phone and upload it directly to our MapBuilder



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OARC Android image collection for mapping

Photos, GPS, sensors in universal format for map creation

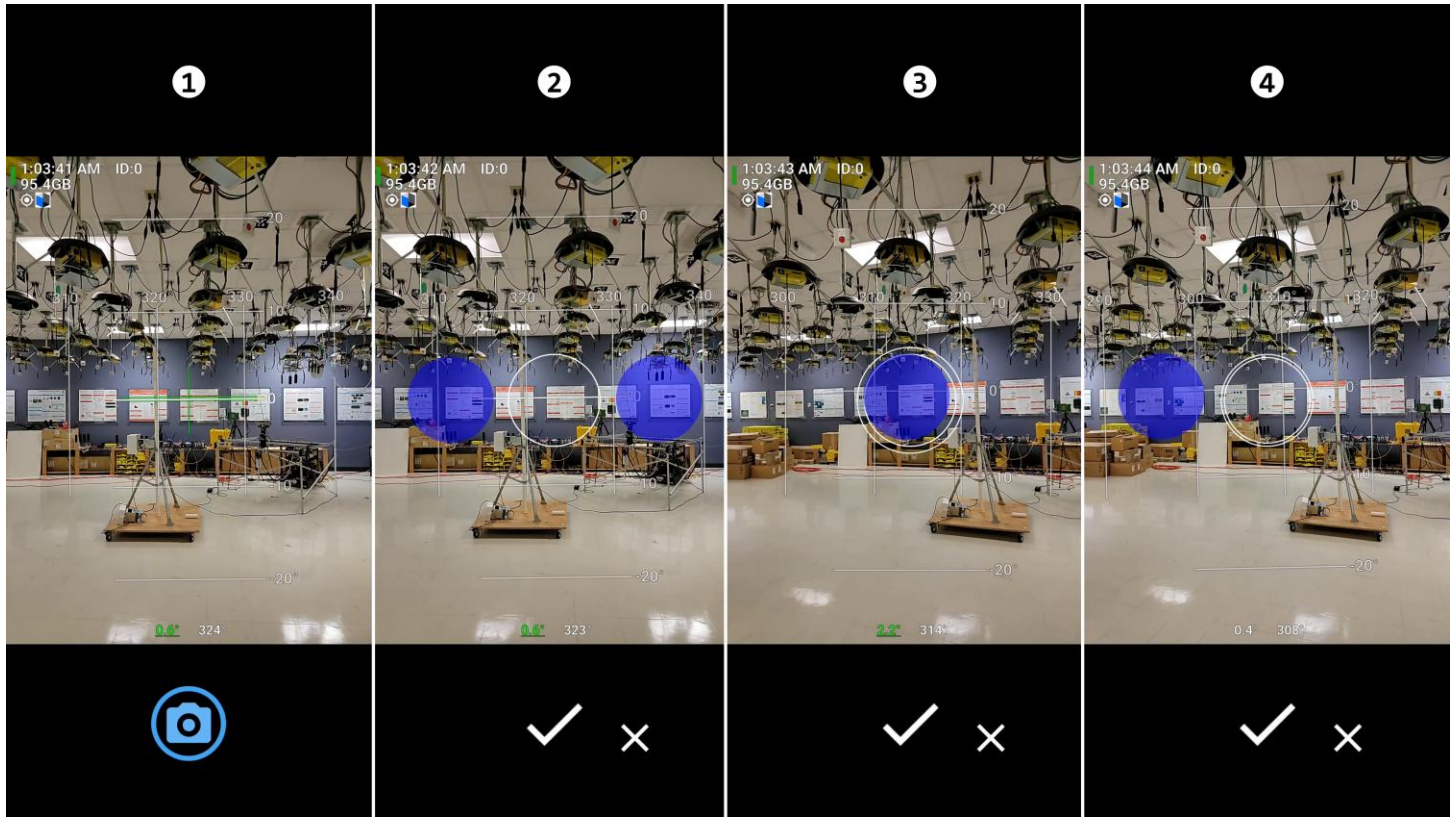
Developed in NGI Atlantic project

Built on OpenCamera

Saving in Kapture format

Data directly applicable in common structure from motion pipelines (e.g. colmap, hloc, etc.)

(Code is a bit outdated, not yet integrated with OpenVPS)



https://opencamera.org.uk/

https://github.com/nokia/AndroidOpenCameraKapture

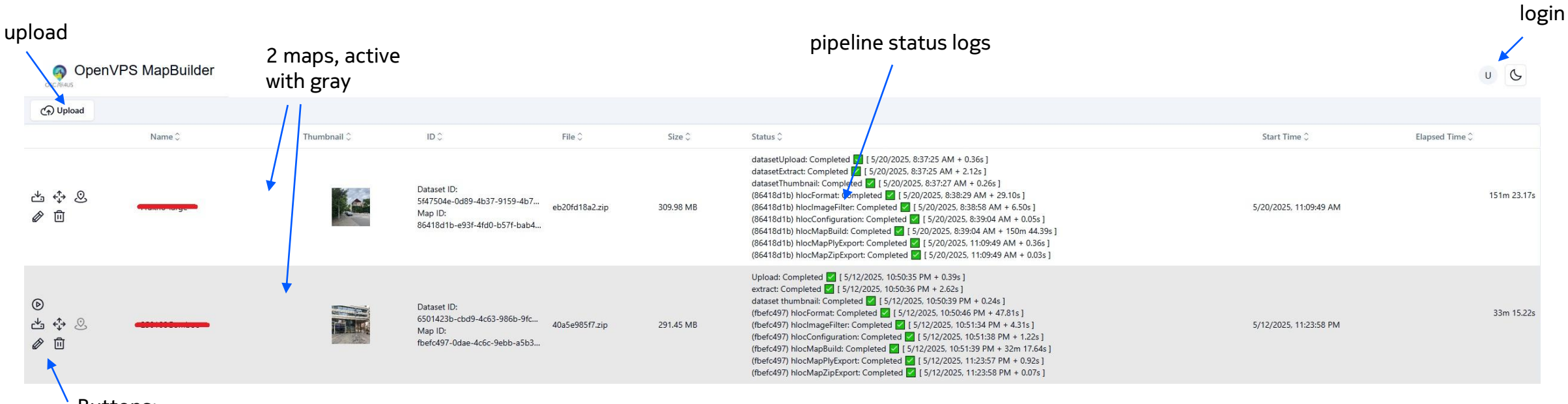
https://europe.naverlabs.com/research/3d-vision/kapture/

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MapBuilder

- Web GUI, iPhone can directly upload a zipped StrayScanner recording
- Automatic extraction, image formatting, image selection, thumbnail, etc.
- Runs HLOC mapping pipeline



Buttons:
Start processing
Download PLY
World alignment
Activate in localizer
Rename
Delete

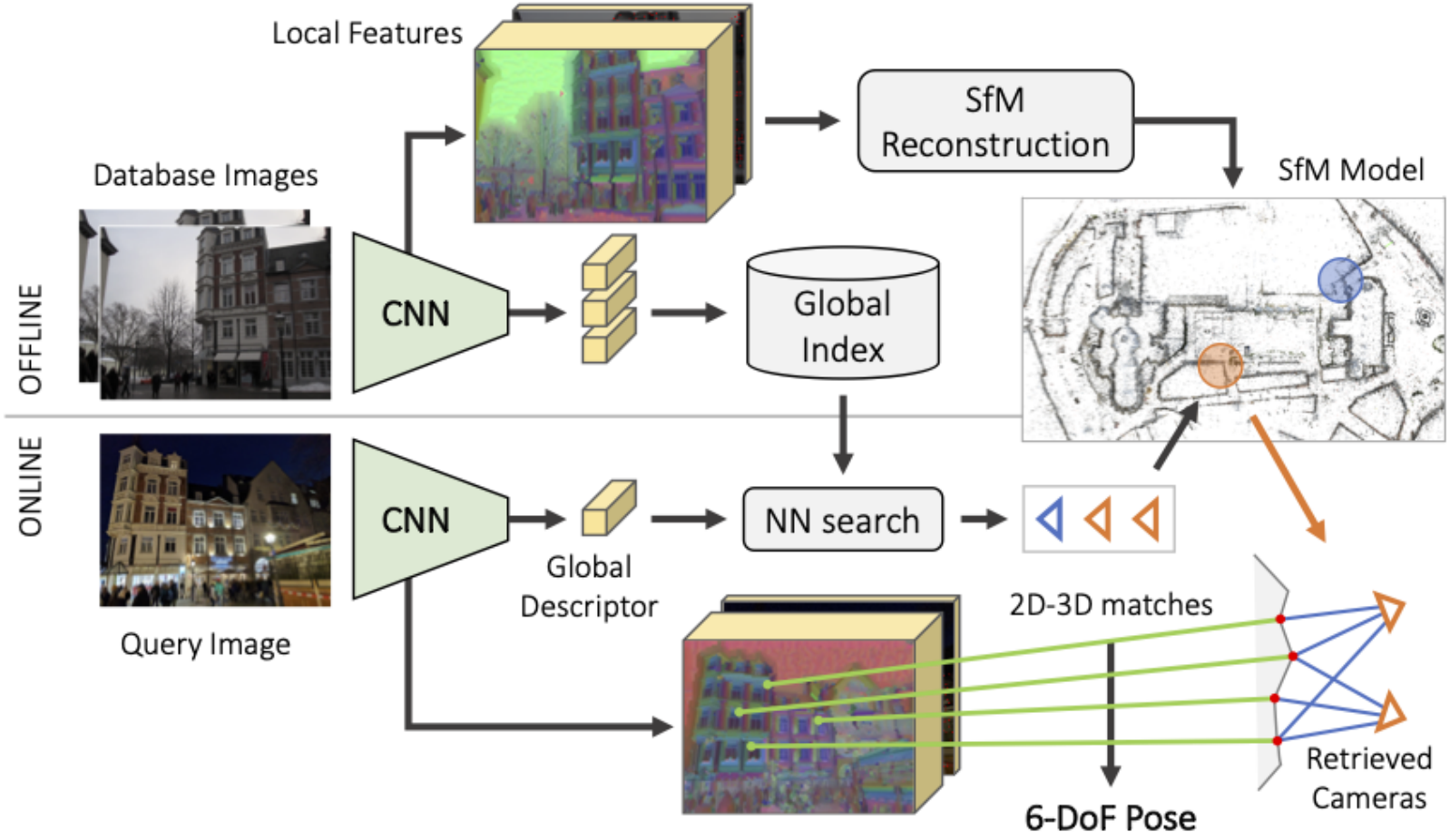
Try it!

https://oarcvps.duckdns.org:8047

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Hierarchical Localization

Image retrieval and feature matching



Sarlin, Cadena, Siegwart, Dymczyk - From Coarse to Fine: Robust Hierarchical Localization at Large Scale (CVPR 2019)

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Hierarchical Localization

ETH Zurich Hierarchical Localization framework

Both mapping and localization

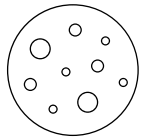
Extensible

Many feature types are already supported

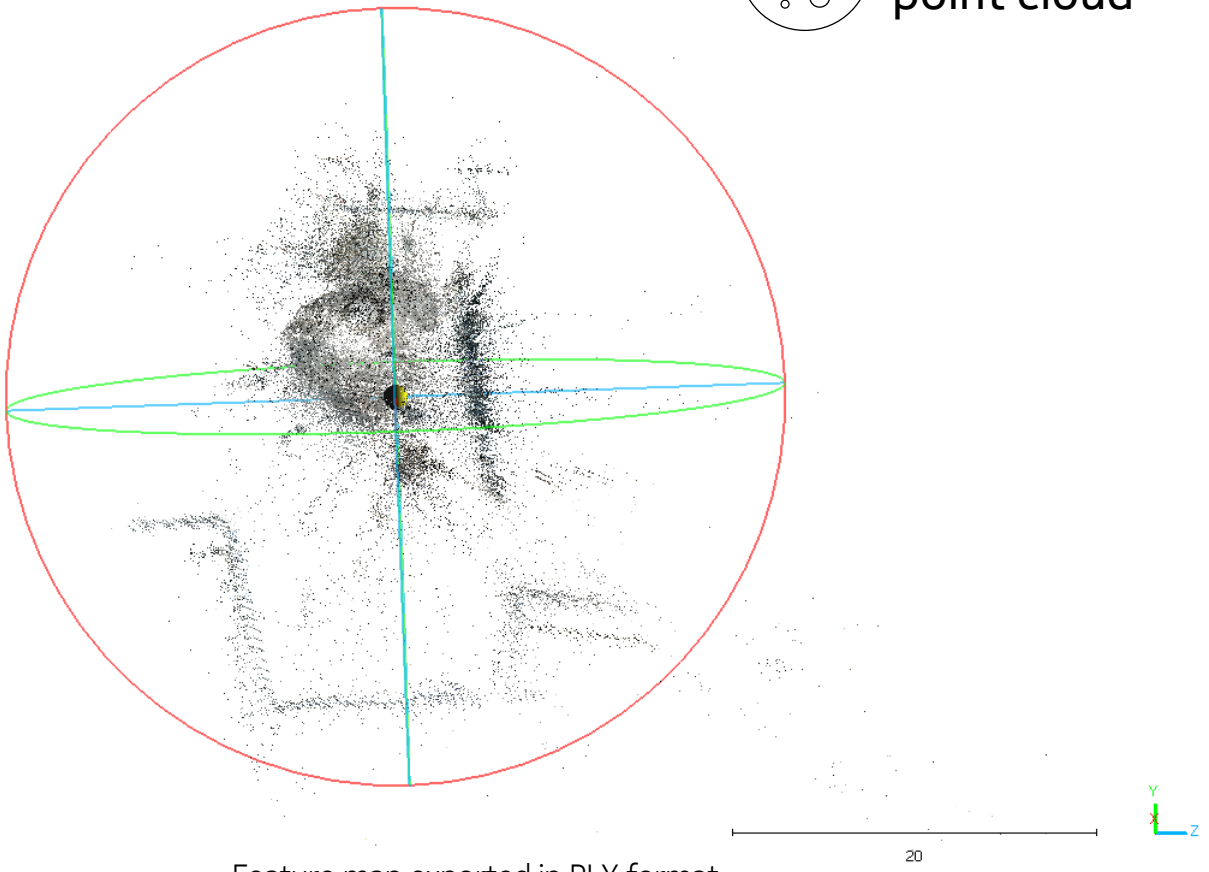
Feature map can be exported in PLY 3D point cloud format for visualization

Built for benchmarking, so a lot of refactoring was needed to make it work with live queries.

`https://github.com/cvg/Hierarchical-Localization`



feature map point cloud



Feature map exported in PLY format

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MapAligner

- Web GUI
- Load OpenStreetMap building footprints as background
- Load PLY point cloud and align it with the building shapes (manual step)

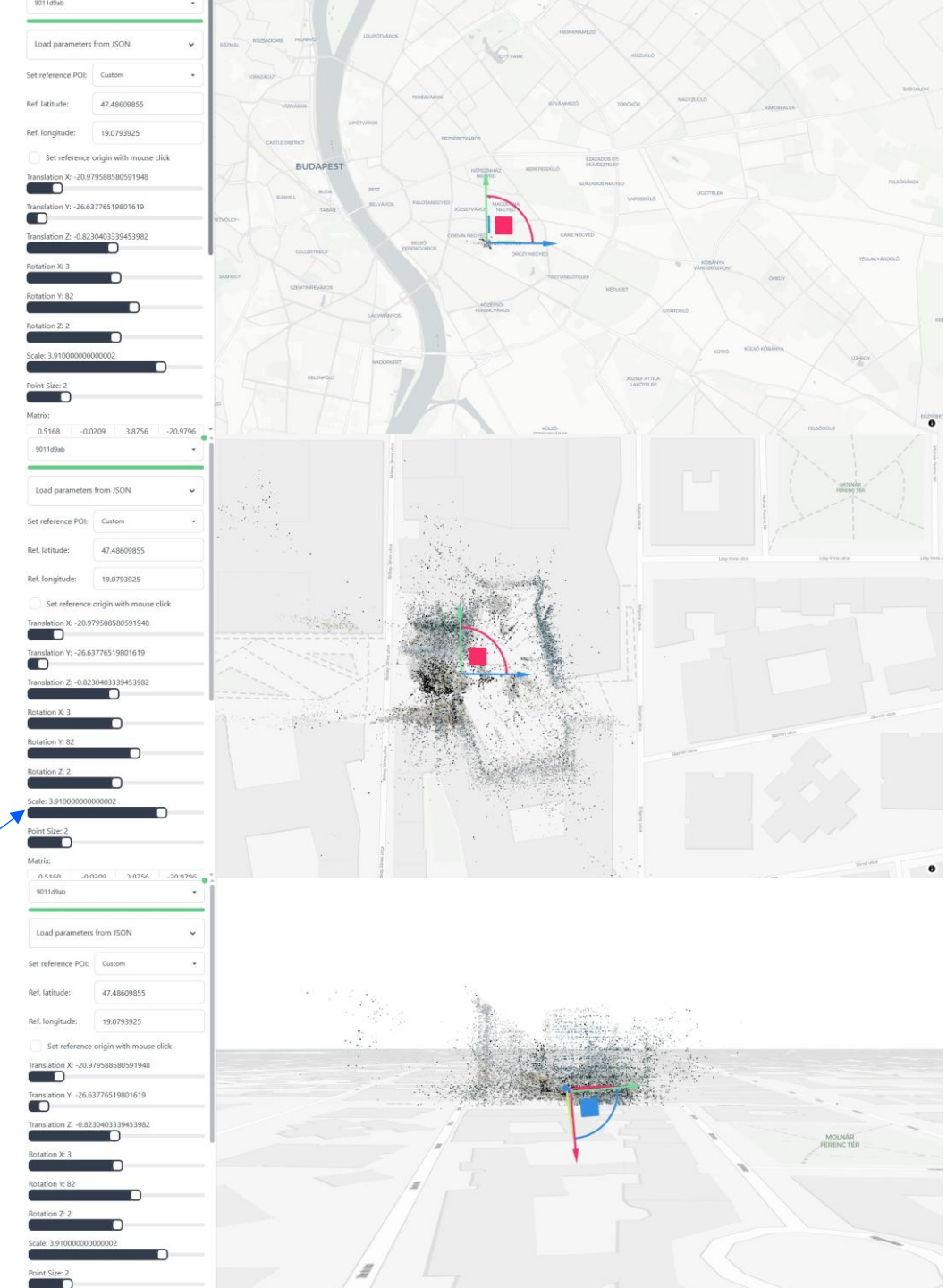
→ geo-alignment transform

- Write resulting transform back to MapBuilder

Scale adjustment is also necessary

Try it!

https://oarcvps.duckdns.org:8048



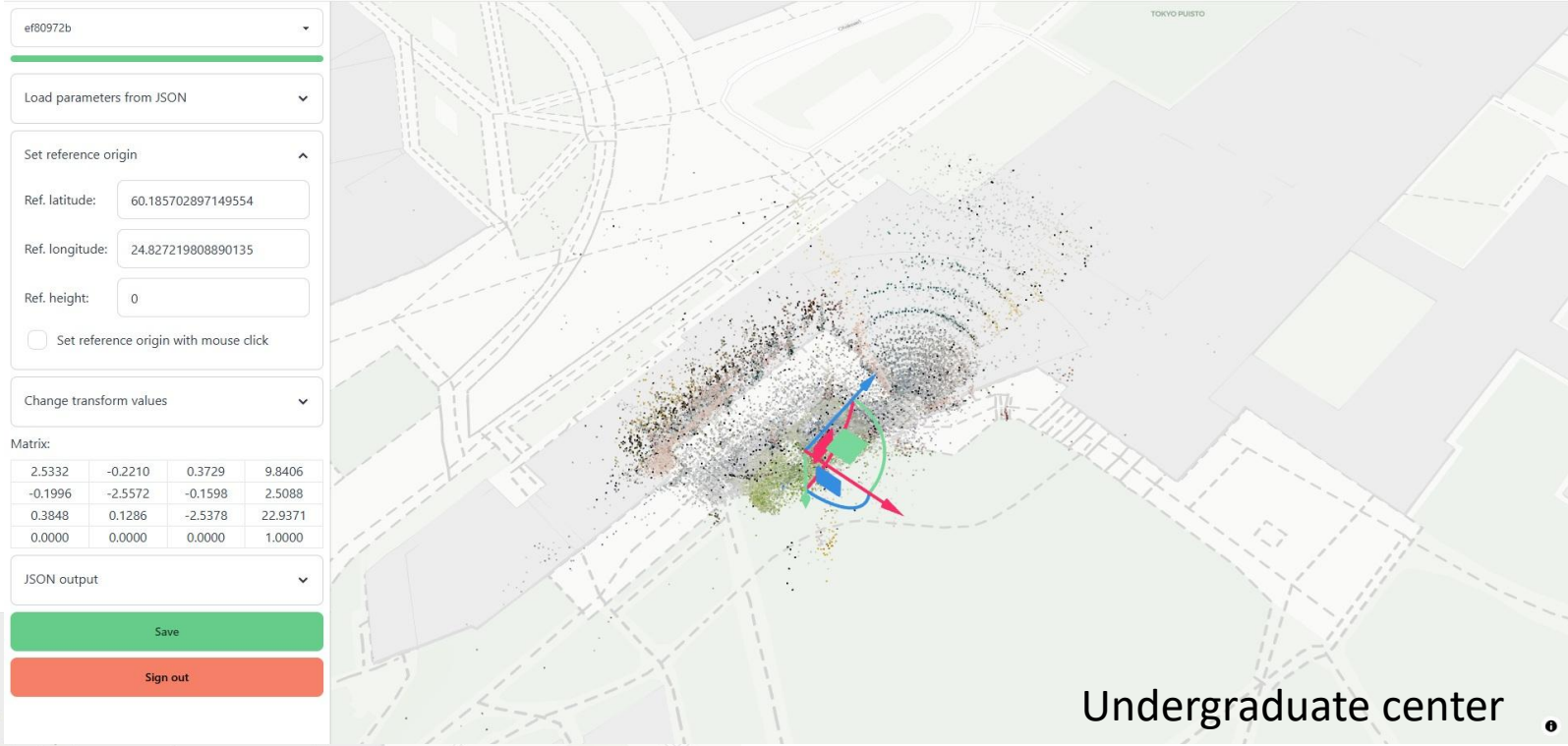
distant view

top view

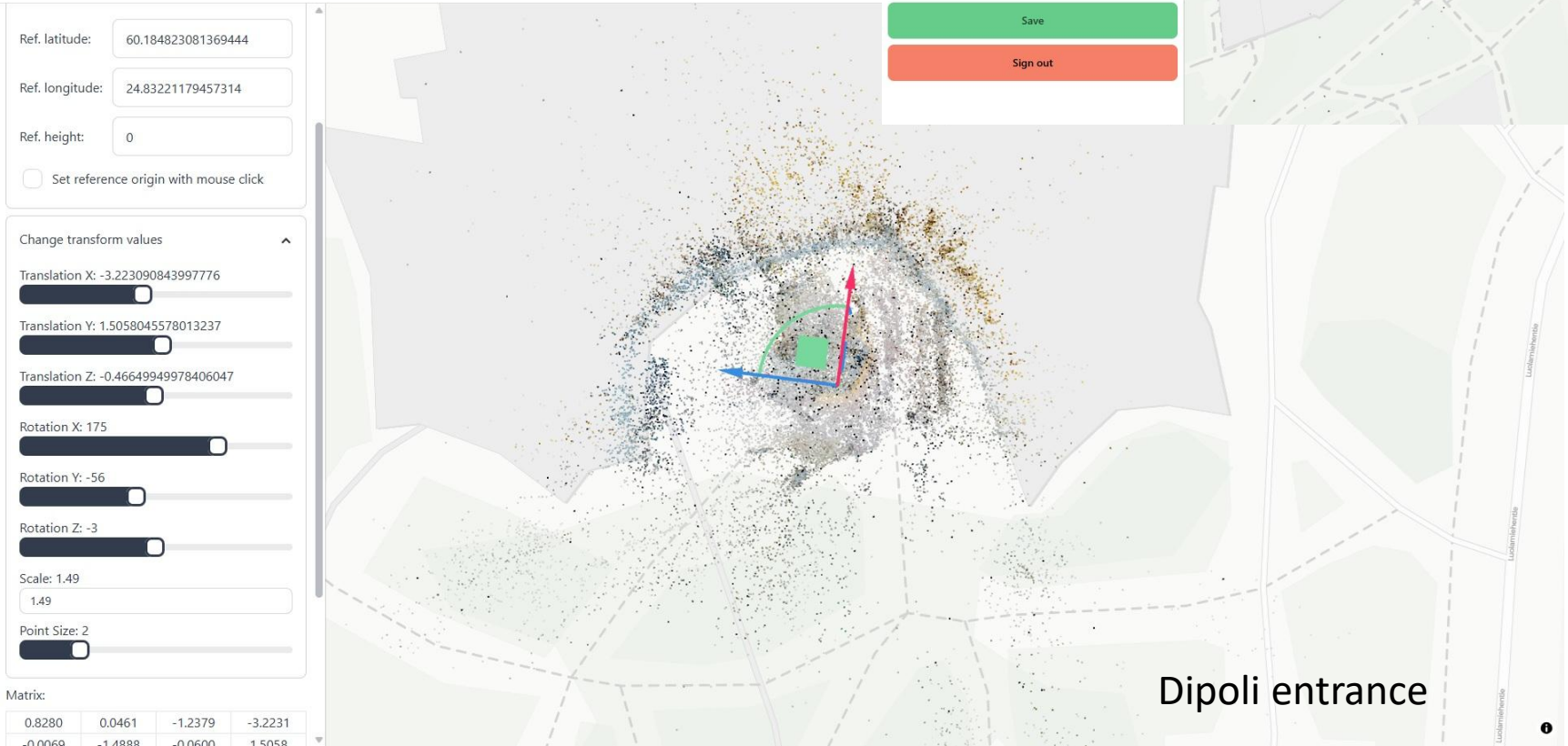
side view

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OpenVPS maps for the tutorial



Undergraduate center



Dipoli entrance

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MapLocalizer



From image to GeoPose…

`https://oarcvps.duckdns.org:8046/localize/geopose`

Open AR Cloud's GeoPoseProtocol

can receive a **GeoPoseRequest** query, sends back **GeoPoseResponse**

```
oscp-geopose-protocol\python>
python.exe demo_client.py \
    --url https://YOUR_VPS_URL/localize/geopose \
    --image ..\data\seattle.jpg \
    --camera ..\data\seattle_camera_params.json \
    --geolocation ..\data\seattle_geolocation_params.json
```

A feature map can be loaded dynamically via REST call by UUID

Current limitation: Only 1 map can be loaded at a time. The active map can be configured in the MapBuilder list. In the future, one could extend the system with a MapSelector component that automatically determines which map(s) to search for a given query (according to coarse estimate in the request)

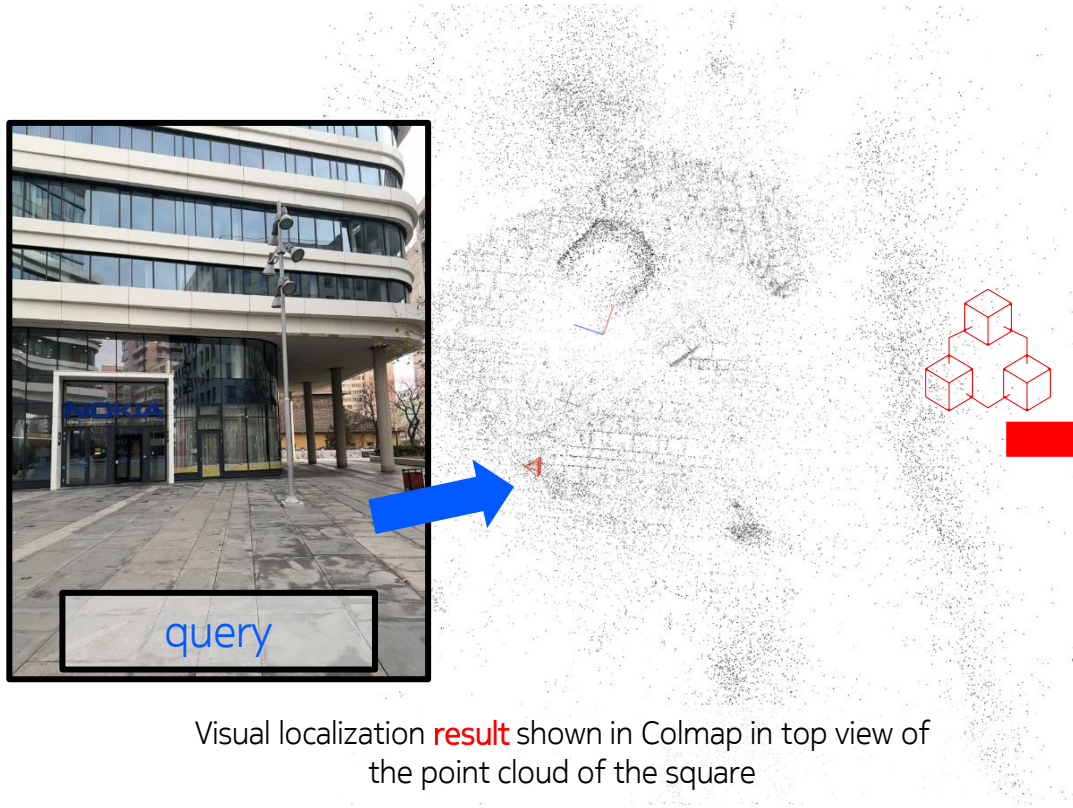
Localization via (refactored) HLOC

Conversion from point cloud pose to GeoPose with the help of the MapAligner's transform.

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Test Query



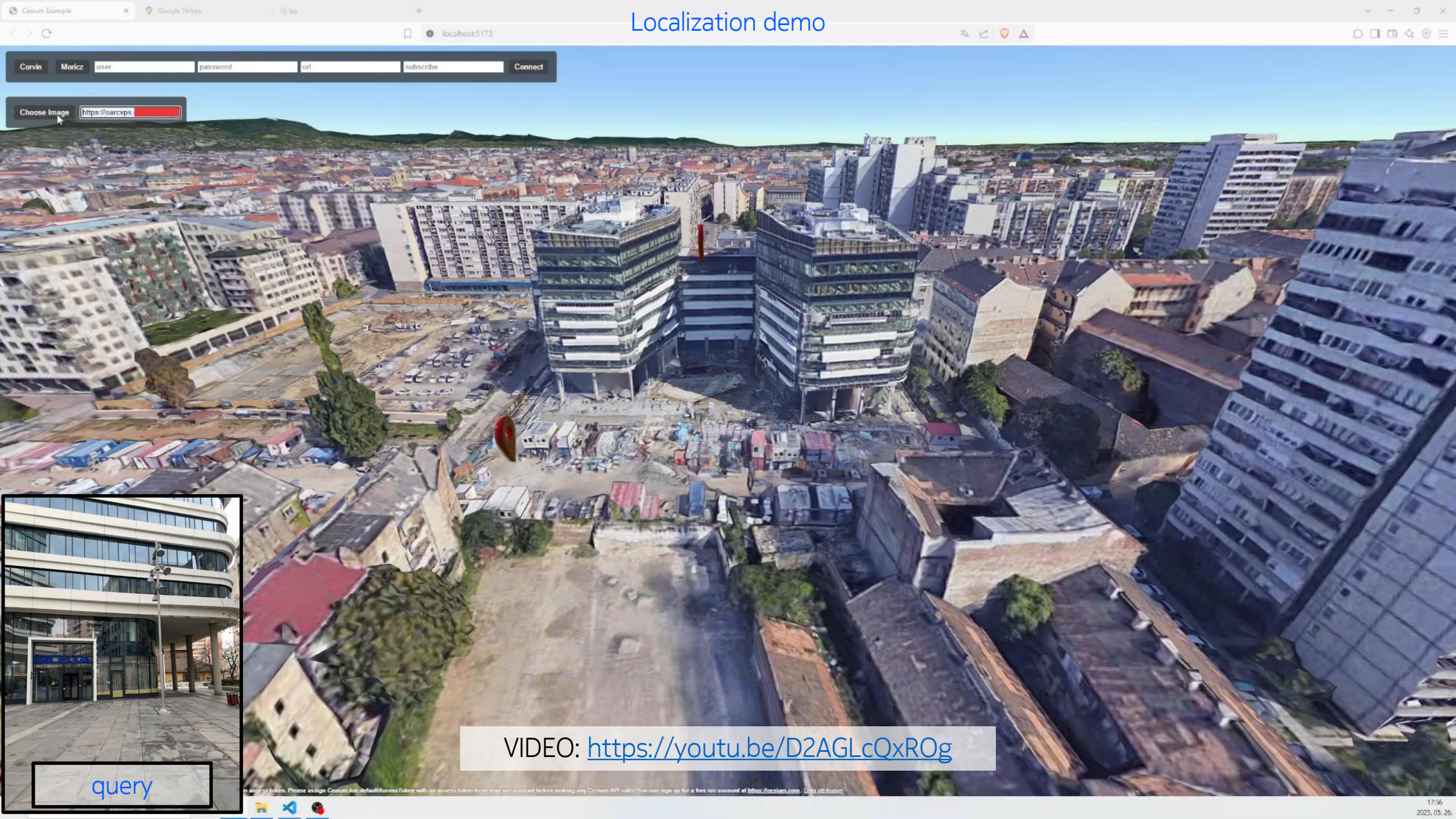
Visual localization **result** shown in Colmap in top view of the point cloud of the square



Visual geo-localization result shown in Globe view:
camera is put into the same WGS84 location and ENU orientation

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Localization demo



Other features

- Authentication
  - Maps contain sensitive data and must be protected
  - We integrated FusionAuth into the system
  - User group management is still missing

- Deployment
  - We dockerized every component and the whole service can be deployed via docker compose
  - Currently running at my home (please do not DDoS ☺), we plan to deploy it on AWS

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Next steps

This is a research tool, not a real service

MapBuilder

- HLOC pipeline configuration from GUI (select feature types, etc.)
- Authentication with user groups, map visibility within groups
- Long list of localization research issues: lighting changes, scene changes, scalability, etc.

MapAligner

- Map alignment initialization from GPS (StrayScanner does not capture)
- Map scale initialization from GPS and/or ARKit poses

MapLocalizer

- Multiple localizer nodes, serving many maps in parallel, dynamic loading and unloading
- Map selector based on prior information (GPS, previous pose, etc.)

WebXR client:

- Switching between services while the user is moving across service boundaries…

Pull requests are welcome: https://github.com/openarcloud/openvps

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Next: Professional VPS applications: Immersal (FI) and Augmented City (IT)

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